Neural Mass Modeling: An Exploration of Population Dynamics and Emergent Behavior Through Reduction of Jansen-Rit

Catherine Martin

April 2nd, 2026

1 Introduction

Neural Mass Modeling captures interaction dynamics between neural populations that are not visible at the scale of single neuron models like the Leaky-Integrate-and-Fire and Hodgkin-Huxley. By zooming out the spatial scale, biological detail is sacrificed for insight into population connectivity. In this work, the Jansen-Rit model is used to examine how excitatory–inhibitory interactions give rise to emergent attractors in phase-plane space. To enable this analysis, the model is reduced to a representative two-dimensional subsystem.

Simulations are used to study how population coupling and parameters shape system behavior, including the emergence of fixed point and limit cycle attractors. An EEG proxy signal is constructed to relate these dynamics to a macroscopic observable, and spectral analysis is used to distinguish dynamical regimes and characterize oscillation frequency.

2 Jansen-Rit Implementation

Jansen and Rit [1995] propose a neural mass model for a cortical column that details interaction dynamics between pyramidal, excitatory interneuron, and inhibitory interneuron subpopulations. The model captures change through second-order differential equations that account for population coupling strength, synaptic delays, and the inertial quality of average membrane potential movement induced by the second-order dynamics. The equations used in this implementation are taken from Jansen and Rit's 1995 work.

$$\begin{aligned}
\dot{y}_0(t) &= y_3(t), \\
\dot{y}_1(t) &= y_4(t), \\
\dot{y}_2(t) &= y_5(t), \\
\dot{y}_3(t) &= Aa\, S(y_1(t) - y_2(t)) - 2a\, y_3(t) - a^2 y_0(t), \\
\dot{y}_4(t) &= Aa\, [p(t) + C_2\, S(C_1 y_0(t))] - 2a\, y_4(t) - a^2 y_1(t), \\
\dot{y}_5(t) &= Bb\, C_4\, S(C_3 y_0(t)) - 2b\, y_5(t) - b^2 y_2(t).
\end{aligned} \tag{1}$$

$$S(v) = \frac{2e_0}{1 + \exp\big(r(v_0 - v)\big)} \tag{2}$$

The six differential equations encode a recurrent feedback loop connecting the subpopulations of the cortical column. The pyramidal cells are driven by net feedback from excitatory and inhibitory interneurons; excitatory interneurons are driven by external input and pyramidal cell activity, while

inhibitory interneurons are driven by pyramidal cell activity alone. If parameters such as coupling strength, external drive, and the delay between excitation and inhibition are tuned appropriately, the model can move into an oscillatory regime in which the observable output rises and falls in response to competing excitation and inhibition. Parameters are described as follows:

- A, B: synaptic gains (excitatory, inhibitory)
- a, b: inverse time constants
- C_1, C_2, C_3, C_4: connectivity constants
- $p(t)$: external input
- $S(\cdot)$: sigmoid firing response

Buzsáki et al. [2012] show that EEG signals arise mainly from postsynaptic currents measured in the extracellular medium. Accordingly, the observable output of the Jansen-Rit model is typically taken to be the net postsynaptic activity of pyramidal cells, reflecting the combined effect of excitatory and inhibitory PSPs. The difference between excitatory and inhibitory interneuronal input to the pyramidal cells, taken as $y_1 - y_2$, is therefore plotted to visualize the oscillatory output (Fig. 1). These oscillations emerge from delayed negative feedback acting on a damped second order system. Concurrent excitatory and inhibitory activation levels are superimposed to aid visualization of the temporal dynamics of the system. Inhibition lags excitation in phase, supporting oscillatory behavior, and each affects the other via pyramidal cell feedback.

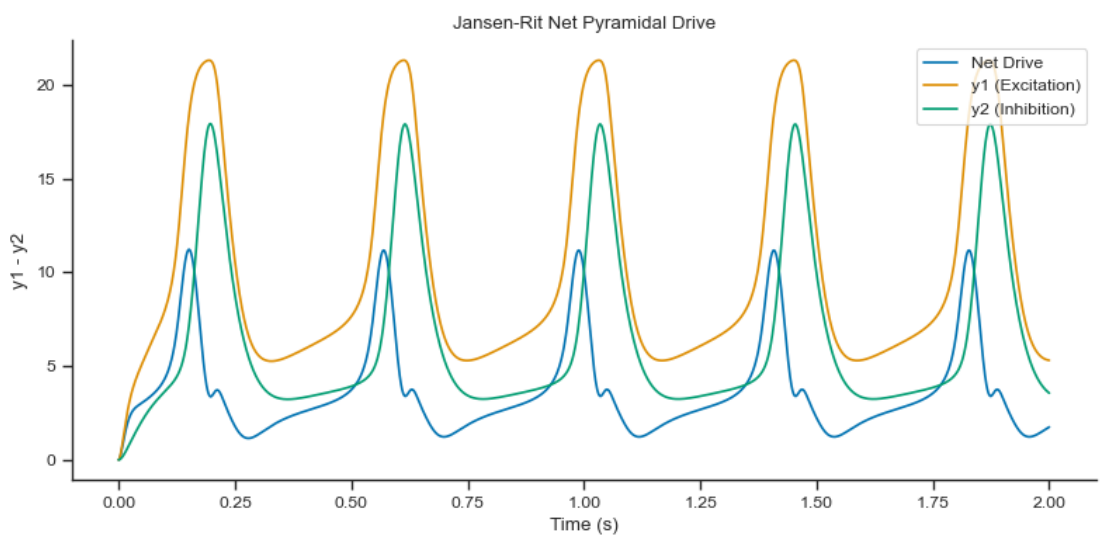


Figure 1: Net Drive to Pyramidal Cells, Excitatory and Inhibitory Interneuron activity curves.

3 Reduction Method

To extend the analysis of population dynamics into phase-plane space, it is advantageous to obtain a representative two-dimensional subsystem of the Jansen-Rit model. Wilson and Cowan [1972] demonstrate that neural population dynamics can be described using coupled excitatory–inhibitory (E–I) systems, which are inherently two-dimensional. Here, we construct a simplified system that retains the E–I feedback structure.

To reduce the Jansen-Rit cortical column to a 2D E–I coupled model, we employ population averaging and phenomenological simplification of the synaptic dynamics into rate-based interactions. The pyramidal–excitatory feedback loop in the original model is collapsed into a single excitatory variable, $E(t)$, representing aggregate excitatory drive, while inhibitory activity is represented by $I(t)$. Second-order synaptic dynamics are approximated by first-order time constants, τ_E and τ_I, since phase-plane analysis does not require detailed synaptic transients for qualitative analysis of dynamical regimes. The connectivity constants are absorbed into a single gain parameter g, representing the effective strength of population coupling. The sigmoid function is used to map average input activity to firing rate. These simplifications yield a coarse-grained Wilson-Cowan-type system that can be expressed as two coupled first-order differential equations with a nonlinear response function:

$$\begin{aligned} \tau_E \frac{dE}{dt} &= -E + S(g(E - I + P_E)) \\ \tau_I \frac{dI}{dt} &= -I + S(g(E + P_I)) \\ S(x) &= \frac{1}{1 + \exp[-a(x - \theta)]} \end{aligned} \tag{3}$$

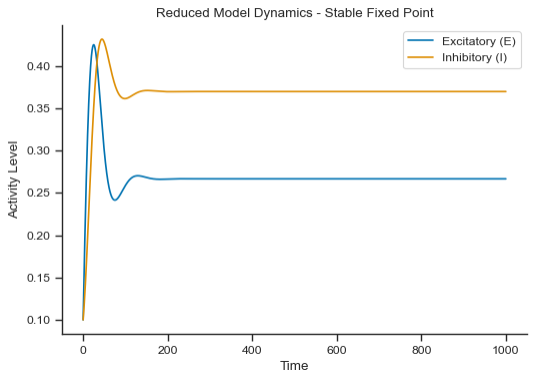
The reduction is reasonable because analysis of dynamical regimes is concerned with the interplay between overall population activity, rather than voltage waveform nuances or synaptic delay layering. Such details do not fundamentally affect the system's ability to create and maintain feedback loops and resulting dynamical phenomena. The core structure responsible for these phenomena is the excitatory-inhibitory feedback interaction, which is preserved in the reduction.

While it retains the critical mechanism, the reduction sacrifices biological detail—the separation between pyramidal and excitatory interneuron populations is lost, along with second-order synaptic transformation and richer waveform structure. Membrane potential–based population states are replaced with variables representing average activity levels. Collapsing the Jansen-Rit connectivity constants into a single gain parameter also removes differentiation in pathway-specific coupling

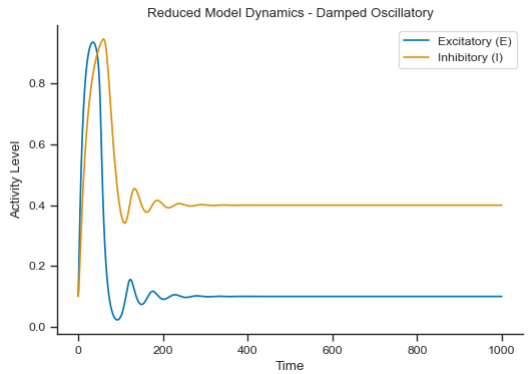
strengths, which is not required to demonstrate dynamical regimes but may be important for studies involving heterogeneous connectivity.

4 Analysis

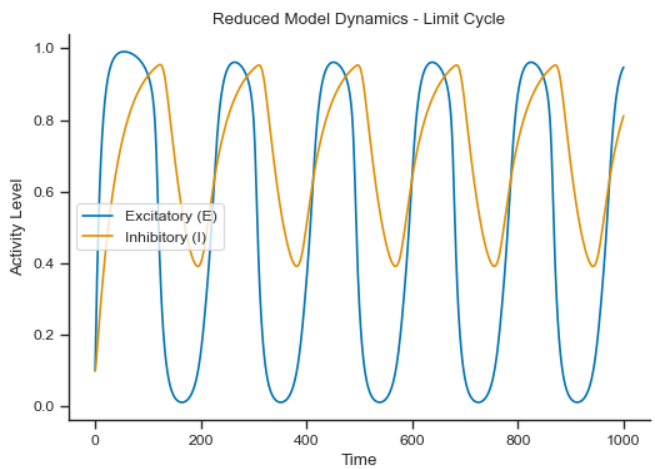
4.1 Dynamical Regimes & Population Interaction



(a) Stable fixed point



(b) Damped oscillation



(c) Limit cycle

Figure 2: Population interaction across regimes.

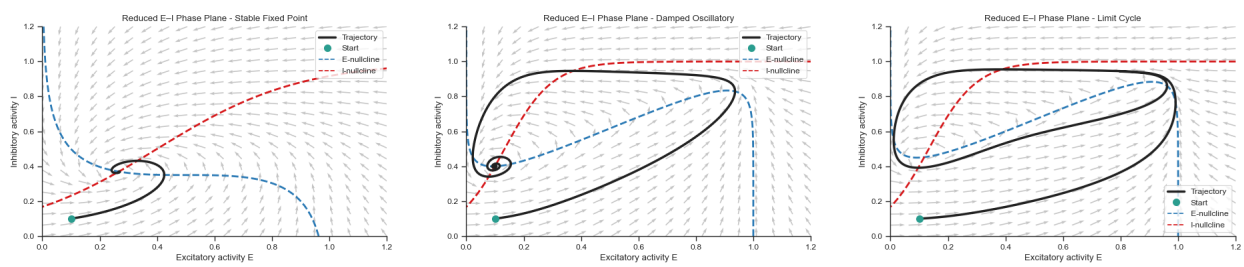
In analyzing the interaction between excitatory and inhibitory populations, it is useful to separate system states into 'regimes' for reference. A stable regime involves attraction of E and I activity levels to a steady value where they remain (Fig. 2a). In this model, a limit cycle or oscillatory regime emerges when feedback between excitation and inhibition is sufficiently strong and temporally lagged to sustain oscillations (Fig. 2c); more generally, such behavior has been shown to arise in excitatory–inhibitory population models where coupling strength and temporal dynamics jointly

determine system stability and oscillatory behavior [Wilson and Cowan, 1972, Kaslik et al., 2022]. The damped-oscillatory state (Fig. 2b) is not strictly a regime but a transition between stable and oscillatory regimes, and will be referred to as a regime here for clarity.

The behavior in these regimes can be obtained through tuning of three parameters of the reduced model: external drive (P_E), inhibitory time constant (τ_I), and gain (g). Each plays a mechanistic role—the external drive sustains excitatory activity, the inhibitory time constant introduces a lag in inhibition, and the gain amplifies feedback between populations.

4.2 Phase Plane Analysis

To better visualize dynamical behavior and transitions between regimes, we analyze the system in phase-plane space rather than explicitly in time. Because nullclines and phase-plane geometry change with parameter variation, it is not feasible to show the full range of geometries produced by parameter sweeps. Fig. 3 shows representative phase-plane behavior corresponding to the activity of regimes in Fig. 2 and the parameter sets that produce them.



(a) Stable fixed point (b) Damped oscillation (c) Limit cycle

Figure 3: Phase-plane representations of dynamical behavior.

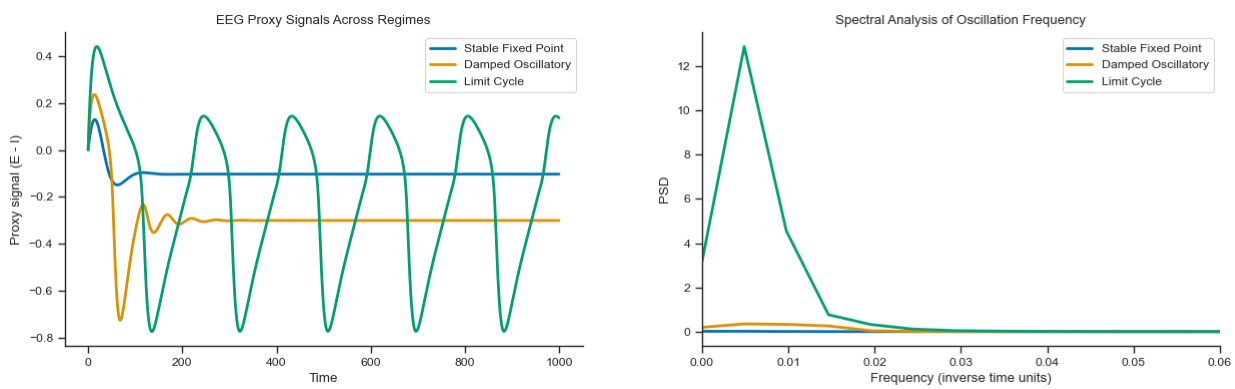
Fig. 3a shows a stable fixed point at the intersection of nullclines, to which the trajectory is attracted without oscillation. Fig. 3b shows a steepening of both the E and I nullclines, shifting their intersection and thus the fixed point to the left. As the gap between the nullclines closes in the top right of the figure, the trajectory moves further into this region but ultimately spirals back to the fixed point; these decreasing spirals correspond to the damped oscillations seen in Fig. 2b. In Fig. 3c, the nullclines shift such that the original fixed point no longer attracts the trajectory, which is guided into a pear-shaped limit cycle. The non-circular nature of the loop reflects the fact that fluctuations of E and I are not perfectly sinusoidal, as a result of sigmoid nonlinearity and the asymmetric interactions and time scales of excitatory and inhibitory populations. For this parameter set, trajectories exhibit sensitivity to initial conditions, with some converging to the fixed point and others to the oscillatory orbit. This suggests the possible presence of multiple attractors and a separating boundary in phase space, though a full characterization of this structure is beyond the scope of the present

analysis. Visuals of these other trajectories are omitted for clarity in the report, and included via demonstration in the project files.

To push the system from a stable regime into a damped-oscillatory regime, the gain parameter was increased from 1.0 to 3.0, reflecting an increase in effective coupling strength. To transition from the damped-oscillatory regime (Fig. 3b) to the oscillatory regime (Fig. 3c), τ_I was increased from 20.0 to 40.0 and external drive from 0.25 to 0.3, as increasing the inhibitory time constant or gain alone did not produce the transition in this parameter regime. This suggests that limit cycle behavior arises only within regions of parameter space where sufficient excitation, delay introduced via time constants, and coupling strength are simultaneously present, highlighting that oscillatory dynamics are not governed by any single parameter but by their joint interaction.

4.3 EEG Proxy, Spectral Analysis, & Parameter Sweeps

To produce a macroscopic EEG-like signal from the simplified model, we take a snapshot of net population activity via $E(t) - I(t)$, representing excitatory–inhibitory balance. This signal reflects oscillatory or stable behavior corresponding to the system state in phase-plane space (Fig. 4a) and provides a simple way to distinguish between regimes.



(a) EEG proxy activity across regimes.

(b) PSD of EEG proxy.

Figure 4: Proxy signal dynamics and spectral content.

Spectral analysis was performed using Welch's method to estimate the power spectral density (PSD) of the EEG proxy signal, describing how signal variance is distributed across frequencies. When stable oscillatory behavior is supported, the PSD shows a dominant low-frequency component near $f \approx 0.005$ (inverse time units), consistent with the long-period ($T \approx 200$ time units) limit cycle observed in the time domain. Power is spread across nearby frequencies due to the slow, non-sinusoidal nature of the oscillation. In contrast, the damped and stable regimes exhibit minimal sustained spectral power, reflecting the absence of persistent oscillations. The absence of a sharp

spectral peak reflects both the slow timescale of the oscillation and the finite observation window used in the analysis. Since the oscillation period is governed by the time required for excitation to build and inhibition to respond, decreases in τ_I directly shorten the oscillatory cycle and would increase its frequency.

These differences in PSD content arise from parameter changes that alter the timing and strength of excitatory-inhibitory feedback. Increasing gain ($g = 1 \rightarrow 3$) produces transient oscillatory behavior, while increasing the inhibitory time constant ($\tau_I = 20 \rightarrow 40$) and external drive ($P_E = 0.25 \rightarrow 0.3$) introduces sufficient delay and sustained excitation to support a stable limit cycle, reflected as increased spectral power in the PSD.

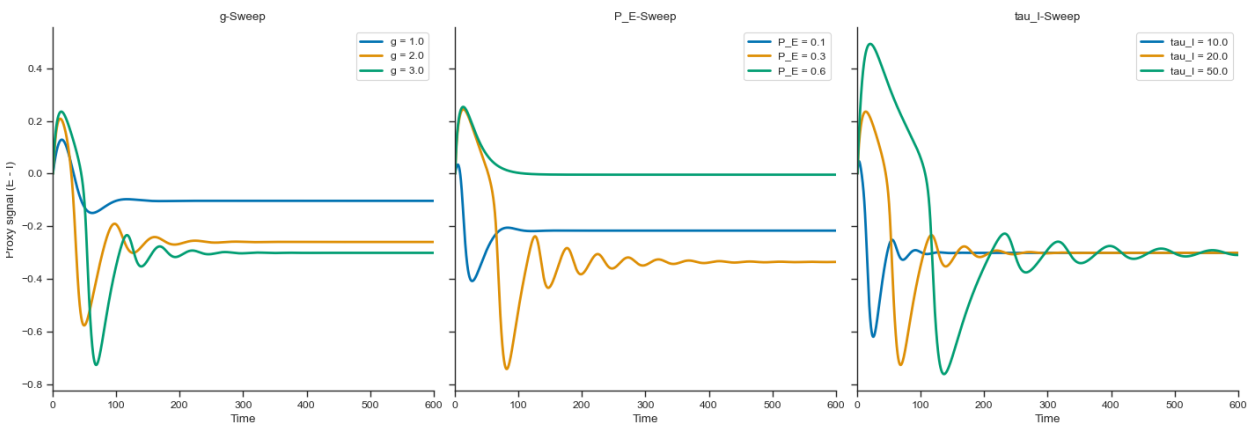


Figure 5: Parameter sweep effects on EEG proxy signal.

The proxy signal also enables parameter sweeps, illustrating how parameters influence model output and regime transitions. Fig. 5 shows sweeps over gain, external drive, and the inhibitory time constant. Higher gain promotes oscillatory behavior in this parameter regime, though with diminishing returns: increasing from $g = 1.0$ to $g = 2.0$ introduces damped oscillations, while increasing from $g = 2.0$ to $g = 3.0$ produces relatively little additional change.

In the P_E sweep, the intermediate value $P_E = 0.3$ produces the strongest oscillatory behavior, indicating that lower external drive does not sufficiently excite the network, while higher levels may reduce oscillations due to saturation. The τ_I sweep shows the strongest oscillations at higher values, suggesting that greater delay between excitation and inhibition promotes oscillatory behavior in this regime. This effect also exhibits diminishing returns, with further increases in τ_I producing no additional qualitative change during further experimentation.

Taken together, the sweeps indicate that a limit cycle attractor for this simplified E–I model appears under a parameter configuration involving high gain, intermediate external drive, and a relatively large inhibitory time constant. The parameter set producing a limit cycle (Fig. 3c) reflects this

structure, with g and τ_I increased significantly from their stable-regime values, while P_E changes more modestly.

5 Limitations & Conclusion

While this reduced Jansen–Rit model enables clearer analysis of dynamical regimes than its six-dimensional parent, the associated tradeoffs should be acknowledged. The reduction abstracts away biophysical detail in favor of tractability, removing distinctions between pyramidal and interneuron populations, simplifying synaptic dynamics to first-order responses, and collapsing heterogeneous coupling into a single effective parameter. As such, the model is best suited for studying population-level interaction dynamics, rather than questions that depend on detailed synaptic structure or pathway-specific connectivity.

Despite these limitations, the reduced model allows direct analysis of the E-I feedback structure that governs system behavior at the mesoscopic scale. In particular, it enables clear interpretation of the mechanisms underlying stable states, oscillatory behavior, and transitions between them. The analysis shows that these behaviors depend on the interaction of three key factors: external drive, inhibitory time scale, and coupling strength. Each contributes a distinct role — setting baseline excitation, introducing temporal delay, and amplifying feedback — and oscillatory dynamics emerge only when these mechanisms act in concert. This reflects the broader principle that feedback systems tend to exhibit oscillatory behavior when sufficient gain and temporal lag combine to sustain repeated overshoot and correction.

A more complete characterization of these transitions would require systematic exploration of parameter space and formal bifurcation analysis, which is beyond the scope of this work. Nonetheless, the present results demonstrate that reduced E–I models capture the essential structure governing qualitative dynamical behavior, and provide a tractable framework for studying how population-level interactions give rise to emergent dynamical phenomena.

6 Implementation Details & Acknowledgments

Simulations are written in python and primarily supported by NumPy and SciPy libraries, the latter of which is used to solve the ODEs in the parent Jansen-Rit model.

This project used generative AI tools as development aids to help clarify concepts, troubleshoot, and refine code and structure. All equations, model behavior, parameter choices, etc. were implemented and validated by the author and represent the author's independent understanding.

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